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Maternal anemia and the risk of preterm birth: a meta-analysis

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ABSTRACT

Background: Globally, preterm birth continues to be a primary contributor to neonatal complications and fatalities. Anemia among the most common nutritional disorders in pregnancy has been proposed as a potential contributor to early delivery. Although extensively studied, the available evidence does not yet provide a clear consensus. This study aimed to conduct a meta-analysis to quantitatively assess the relationship between maternal anemia and the risk of preterm birth.

Methods: This meta-analysis was conducted and reported in accordance with the PRISMA guidelines and the MOOSE checklist. A systematic and exhaustive search was performed across multiple electronic databases PubMed, Scopus, Web of Science, Embase, and the Cochrane Library to identify relevant studies published from inception to 1 January 2025. Effect sizes were combined using a random-effects meta-analysis.

Results: A total of 45 articles, reporting 60 independent study populations comprising 2,119,392 pregnant individuals were included. Considerable heterogeneity was found across the included studies ($I^2 = 95.54\%$, $p < 0.001$), which justified the application of a random-effects model for pooling effect sizes. Maternal anemia was significantly associated with an increased risk of preterm birth (pooled odds ratio [OR] = 1.28, 95% confidence interval [CI]: 1.20–1.36, $p < 0.001$). Despite substantial heterogeneity, sensitivity analyses confirmed the robustness of this association. The relationship was strongest in studies conducted in Asia (OR = 1.30, 95% CI: 1.22–1.40; $p < 0.001$) and the Europe (OR = 1.24, 95% CI: 1.08–1.41; $p = 0.001$) and reached statistical significance when anemia was assessed during the first trimester (OR = 1.12, 95% CI: 1.03–1.51; $p = 0.007$) and the third trimester (OR = 1.65, 95% CI: 1.42–1.91; $p < 0.001$), while no significant associations were found in the second trimesters (OR = 1.10, 95% CI: 0.99–1.22; $p = 0.05$). Funnel plot asymmetry and a significant Egger's test ($p = 0.001$) indicated potential publication bias, although Begg's test was not significant ($p = 0.425$).

Conclusions: The current evidence suggests that maternal anemia, particularly in the third trimester, is significantly associated with an increased risk of preterm birth. These findings emphasize the clinical imperative for comprehensive and timely anemia screening during the third trimester. Integrating targeted interventions such as iron and micronutrient supplementation, is essential to mitigate the risk of preterm delivery and improve neonatal outcomes.

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Maternal anemia; preterm birth; pregnancy; meta-analysis; neonatal outcomes

1. Introduction

Preterm birth defined as childbirth occurring before 37 completed weeks of gestation remains a major global health concern, significantly contributing to neonatal mortality, long-term morbidity, and healthcare system strain worldwide. Despite advancements in obstetric care, preterm birth rates remain persistently high, particularly in low- and middle-income countries (LMICs) where access to comprehensive prenatal services is often limited. Identifying and addressing modifiable risk factors is vital to mitigating both the incidence and the negative consequences associated with preterm birth [1].

Among the various contributing factors, maternal nutritional status (particularly iron levels) has garnered considerable attention due to its high global prevalence and potential for preventive

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intervention. Anemia, particularly that stemming from iron deficiency, remains one of the most common nutritional disorder during pregnancy, affecting between 4.5% and 51% of pregnant women worldwide [2]. Maternal iron deficiency is known to contribute to negative pregnancy sequelae, encompassing increased maternal illness, low birth weight, prematurity, and intrauterine growth restriction [3]. From a physiological perspective, anemia can impair oxygen delivery to the placenta and fetus, potentially compromising fetal development and activating biological pathways that may induce preterm labor through inflammatory or hormonal responses [4,5].

Despite widespread awareness about the prevalence of anemia among the maternal cohort (i.e. maternal anemia) [6,7], its association with preterm birth remains inconclusive [8–51]. While several studies suggest a strong correlation, particularly during the first and second trimesters, others have reported no significant relationship [4,52]. These inconsistencies may arise from differences in study designs, definitions of anemia, timing of hemoglobin measurement, population demographics, and adjustment for confounding variables [4].

Meta-analysis constitutes a statistical approach designed to integrate results across numerous studies to arrive at a synthesized conclusion. This method proves especially advantageous when original studies yield contradictory outcomes or possess insufficient statistical power individually. By pooling data from disparate sources, meta-analyses not only bolster statistical power but also sharpen effect estimate precision, thereby enabling a more complete and integrated comprehension of the research topic. Given these discrepancies, a meta-analysis is warranted to synthesize existing evidence, quantify the overall effect, and explore potential sources of heterogeneity [53]. This study aims to provide a robust and updated, comprehensive analysis of the relationship between maternal anemia and preterm birth risk. It includes more recent studies and uses improved methodology to inform both clinical practice and public health policy.

2. Material and methods

2.1. Information sources

This is a secondary study of the meta-analysis type that addresses the relationship between maternal anemia and preterm birth. Adherence to a rigorous, pre-specified methodological plan was maintained throughout this investigation. The protocol detailing the scope, eligibility criteria, and analytical approach for this meta-analysis was formally logged in the PROSPERO International Prospective Register of Systematic Reviews before the commencement of our literature search. This upfront registration provides assurance regarding the study's integrity and non-duplication. (PROSPERO Registration Number: CRD420251129553). This registration guarantees our adherence to the pre-established methodology aligned with the objectives outlined in this study. This review adhered to and was reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [54] and the MOOSE (Meta-analyses Of Observational Studies in Epidemiology) checklist [55].

2.2. Search strategy

A systematic and exhaustive search was performed across multiple electronic databases PubMed, Scopus, Web of Science, Embase, and the Cochrane Library to identify relevant studies published up to 1 January 2026. We designed the search strategy using both relevant keywords and standardized Medical Subject Headings (MeSH) such as “maternal anemia,” “pregnancy anemia,” “preterm birth,” “preterm delivery,” and “risk factors.” Boolean operators (AND, OR) were systematically applied throughout the search process. Additionally, reference lists of relevant studies and prior reviews were manually screened to capture any additional eligible publications.

2.3. Eligibility criteria

Studies were included if they met the following criteria: (1) Observational studies (cohort, case-control, or cross-sectional) investigating the association between maternal anemia during pregnancy and

preterm birth (mothers with anemia as the case group and healthy mothers as the control group); (2) Clearly defined anemia based on hemoglobin thresholds consistent with WHO criteria; (3) Provided sufficient data to extract or calculate effect measures such as odds ratios (ORs), risk ratios (RRs), or hazard ratios (HRs) with 95% confidence intervals (CIs) and (4) Published in English.

Exclusion criteria were as follows: (1) Studies lacking outcome data or providing unclear definitions of anemia or preterm birth; (2) Case reports, reviews, editorials, and conference abstracts without full datasets; and (3) Studies focusing exclusively on high-risk populations with specific comorbidities (e.g. gestational diabetes, preeclampsia) were not included, unless such studies were separately analyzed, as their inclusion might introduce bias into the study results.

2.4. Study selection

Two independent reviewers (e.g. M.A. and M.M.N.) screened titles and abstracts for relevance. Full-text articles of potentially eligible studies were reviewed in detail. Any discrepancies identified were resolved through discussion among the reviewers or by consulting a third independent reviewer.

2.5. Data collection process

Data extraction was conducted independently by two reviewers. Data recorded on paper were compiled and formatted according to the study objectives, and subsequently entered into SPSS version 17.

2.6. Data Items

Extracted data included: author, publication year, country, study design, sample size, timing and definition of anemia, severity levels, criteria for preterm birth, effect estimates (OR, RR, HR), and adjusted confounders.

2.7. Bias assessment

The methodological quality of included studies was independently assessed using the Newcastle-Ottawa Scale (NOS) for observational studies [56]. Risk of bias was also systematically evaluated and documented. If the scores obtained from the NOS checklist for a single study exhibited a significant discrepancy between two evaluators, to the extent that it altered the qualitative classification of the study, a third evaluator performed the qualitative assessment. Subsequently, a group discussion was convened, and the discrepancy was resolved. Studies were categorized into low, moderate, and high quality based on their scores: 0–5 points indicated low quality, 6–7 points moderate quality, and 8–9 points high quality. A minimum score of 6 was required for inclusion in the quantitative meta-analysis, with articles meeting this threshold proceeding to data extraction and meta-analysis.

2.8. Synthesis methods

Effect measures were pooled using a random-effects model (DerSimonian and Laird method) to account for between-study variability [57]. Heterogeneity was assessed using Cochran's Q test and quantified with the I^2 statistic; values exceeding 50% were considered to indicate substantial heterogeneity [58]. Subgroup analyses were conducted based on anemia severity, trimester of anemia diagnosis, geographic location, and study design. We performed sensitivity analysis by excluding one study at a time. We evaluated publication bias using funnel plot symmetry and Egger's regression test. All statistical analyses were performed using Comprehensive Meta-Analysis (version 2). A p-value < 0.05 was considered statistically significant.

3. Results

3.1. Study selection and characteristics

Forty-five articles were ultimately included, which yielded data from 60 separate studies. This discrepancy arose because some of these articles reported findings related to two different populations or analyzed outcomes across three separate trimesters (Figure 1). These studies employed cohort, case-control, and cross-sectional designs, with data collected from regions including Asia, Africa, Europe, and the Americas (Table 1).

3.2. Overall effect of maternal anemia on preterm birth

The combined results indicated a statistically significant increase in the likelihood of preterm birth among mothers diagnosed with anemia during pregnancy (pooled OR = 1.28, 95% confidence interval [CI]: 1.20–1.36, $p < 0.001$) (Figure 2).

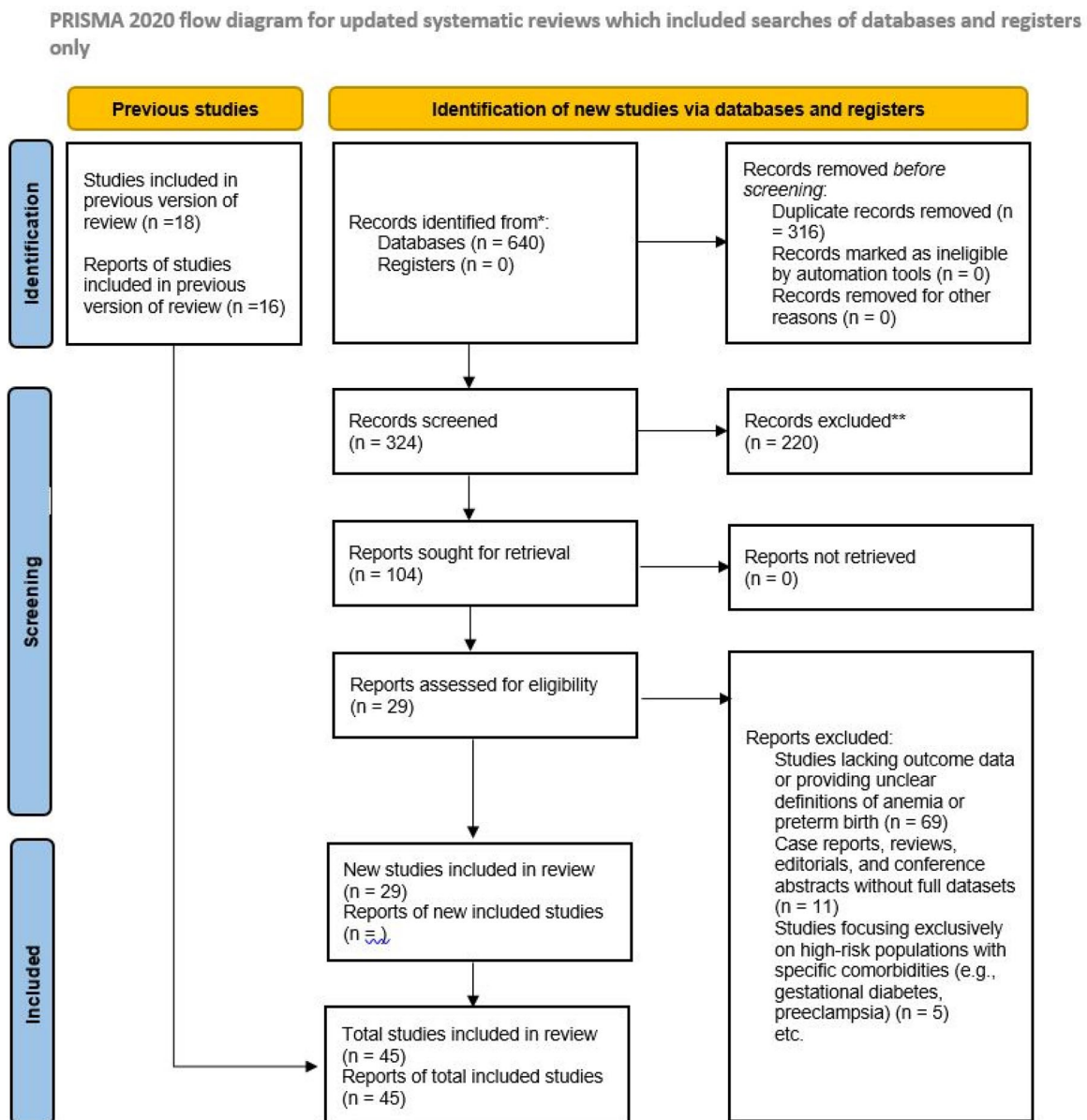


Figure 1. The approach used to determine study eligibility for meta-analysis.

Table 1 . Essential features of the studies included in the meta-analysis.

Ref.	First author, Published Year	Study Type	Place	Trimester	Cut of point		Score of Quality
					Preterm (week)	Anemia (g/L)	
6	Burden CA, 2023 (2nd trimester)	Prospective	United Kingdom	Second	<37	<11	9
6	Burden CA, 2023 (3rd trimester)	Prospective	United Kingdom	Third	<37	<11	9
7	Khezri R, 2023	Retrospective	Iran	First-and-second	<37	<11	8
8	Ardic c, 2019	Retrospective	Türkiye	First-and-second	<37	<11	7
9	Wu L, 2022 (2nd trimester)	Retrospective	China	Second	<37	<11	8
9	Wu L, 2022 (3rd trimester)	Retrospective	China	Third	<37	<11	8
10	Tang L, 2024	Retrospective	China	First	<37	<11	9
11	Kant Sh, 2018	Retrospective	India	Third	<37	<11	7
12	Ibrahim Omar A, 2022	Retrospective	Somalia	Third	<37	<11	8
13	Safarzadeh S, 2023	Retrospective	Iran	Third	<37	<10.5	8
14	Kibret Y, 2024	Retrospective	Ethiopia	Third	<37	<11	7
15	Elmugabil A, 2022	Retrospective	Sudan	Third	<37	<11	7
16	Shah T, 2022	Retrospective	Pakistan	Third	<37	<11	7
17	Heydarpour F, 2019 (1st trimester)	Retrospective	Iran	First	<37	<11	8
17	Heydarpour F, 2019 (3rd trimester)	Retrospective	Iran	Third	<37	<11	8
18	Abadiga M, 2021	Retrospective	Ethiopia	Third	<37	<11	7
19	Salucketch S, 2020	Retrospective	Thailand	Third	<37	<11	8
20	Mekonen D, 2019	Retrospective	Ethiopia	Third	<37	<11	8
21	Sun Ch, 2021 (1st trimester)	Retrospective	China	First	<37	<11	7
21	Sun Ch, 2021 (2nd trimester)	Retrospective	China	Second	<37	<11	8
21	Sun Ch, 2021 (3rd trimester)	Retrospective	China	Third	<37	<11	8
22	Mekuriyaw A, 2020	Retrospective	Ethiopia	Third	<37	<11	8
23	Harrison R, 2021	Retrospective	America	Third	<37	<11	8
24	Yilmaz E, 2018	Retrospective	Turkey	Third	<37	<11	8
25	Ahmad U, 2022	Retrospective	Pakistan	Third	<37	<12	8
26	Iqbal R, 2020	Retrospective	Pakistan	Third	<37	<10	8
27	Aziz S, 2021	Prospective	Pakistan	First	<37	<11	7
28	Mohamed K, 2022	Retrospective	Iraq	Third	<37	<11	7
29	Anwar R, 2019	Retrospective	Pakistan	Third	<37	<11	7
30	Mahmood T, 2019	Retrospective	Pakistan	First-and-second	<37	<10	6
31	Wu Sh, 2024	Retrospective	Taiwan	Third	<37	<11	8
32	Wagura P, 2018	Retrospective	Kenya	Third	<37	<10	8
33	Nsereko E, 2020	Prospective	Rwanda	First	<37	<11	8
34	Bondevik G, 2001	Retrospective	Nepal	First	<36	hematocrit below 34	7
35	Abeyseena C, 2010	Prospective	Sri Lanka	First	<37	<11	8
36	Kidnato HL, 2009	Retrospective	Tanzania	First	<37	<11	8
37	Ren A, 2007	Retrospective	China	First	<37	<11	7
38	Levy A, 2005	Retrospective	Israel	First	<37	<10	9
39	Chang CS, 2003 (2nd trimester)	Retrospective	USA	Second	<37	<10.5	9
39	Chang CS, 2003 (3rd trimester)	Retrospective	USA	Third	<37	<10.5	9
40	Hamalain H, 2003 (1st trimester)	Retrospective	Finland	First	<37	<11	8
40	Hamalain H, 2003 (2nd trimester)	Retrospective	Finland	Second	<37	<11	8
40	Hamalain H, 2003 (3rd trimester)	Retrospective	Finland	Third	<37	<11	8
41	Zhou LM, 1998	Retrospective	China	First	<37	<11	7
42	Knottnerus JA, 1990	Retrospective	Netherland	Third	<37	hematocrit below 38	7
43	Xiong X, 2003 (1st trimester)	Retrospective	China	First	<37	<10	7
43	Xiong X, 2003 (3rd trimester)	Retrospective	China	Third	<37	<10	7
44	Kermani Moakhar H, 2019 (1st trimester)	Retrospective	Iran	First	<37	<10.5	7

(Continued)

Table 1. Continued.

Ref.	First author, Published Year	Study Type	Place	Trimester	Cut of point		Score of Quality
					Preterm (week)	Anemia (g/L)	
44	Kermani Moakhar H, 2019 (2nd trimester)	Retrospective	Iran	Second	<37	<10.5	7
44	Kermani Moakhar H, 2019 (3rd trimester)	Retrospective	Iran	Third	<37	<10.5	6
45	Gonzales GF, 2012	Retrospective	Peru	All	<37	<11	9
46	Huang LL, 2015 (1st trimester)	Retrospective	China	First	<37	<11	9
46	Huang LL, 2015 (2nd trimester)	Retrospective	China	Second	<37	<11	9
46	Huang LL, 2015 (3rd trimester)	Retrospective	China	Third	<37	<11	9
47	Alwan NA, 2015 (1st & 2nd trimester)	Retrospective	UK	First-and-second	<37	<11	9
47	Alwan NA, 2015 (2nd & 3rd trimester)	Retrospective	UK	Second-and-third	<37	<10.5	9
48	Pakniat H, 2016	Retrospective	Iran	First	<37	<11	6
49	Zhang Y, 2017 (1st trimester)	Retrospective	China	First	<37	<11	9
49	Zhang Y, 2017 (3rd trimester)	Retrospective	China	Second	<36	<11	9
50	Hajian K, 2005	Retrospective	Iran	Third	<37	<11	7

3.3. Subgroup analyses

3.3.1. By geographic region

When analyzed by continent, the relationship between maternal anemia and preterm birth was most pronounced in studies from Asia (OR = 1.30, 95% CI: 1.22–1.40; $p < 0.001$) and Europe (OR = 1.24, 95% CI: 1.08–1.41; $p = 0.001$), compared to those conducted in America (OR = 0.91, 95% CI: 0.64–1.30; $p = 0.62$) and Africa (OR = 1.98, 95% CI: 0.89–4.40; $p = 0.91$). The difference between them was not significant ($p = 0.159$) (Table 2).

3.3.2. By trimester of anemia assessment

A statistically significant association between maternal anemia and preterm birth was observed in first trimester (OR = 1.12, 95% CI: 1.03–1.51; $p = 0.007$) and the third trimester (OR = 1.65, 95% CI: 1.42–1.91; $p < 0.001$). In contrast, no significant associations were found in the second trimester (OR = 1.10, 95% CI: 0.99–1.22; $p = 0.05$) (Table 2).

3.3.3. By study design

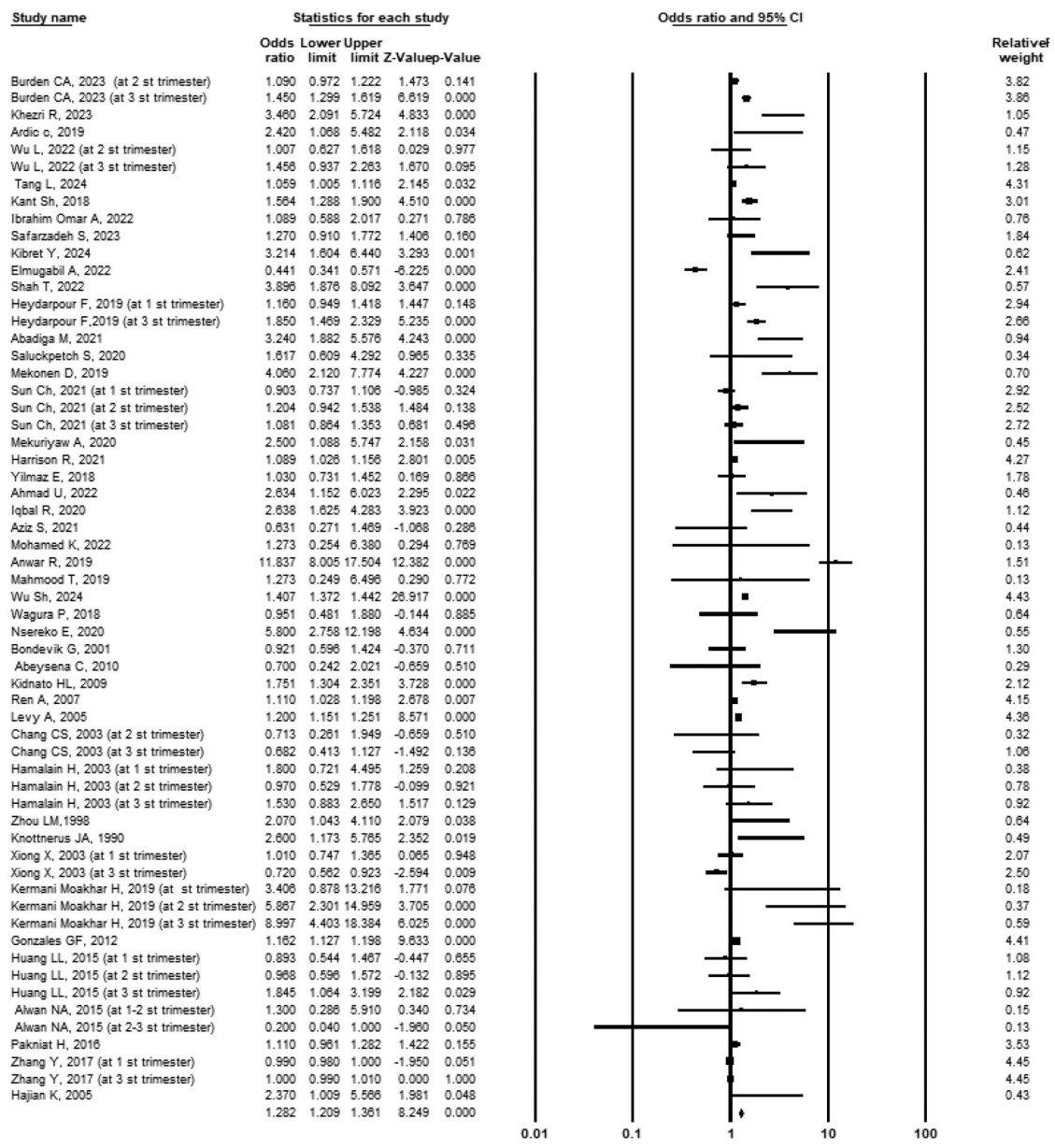
Retrospective studies demonstrated a statistically significant association, with a pooled OR of 1.27 (95% CI: 1.20–1.35, $p < 0.001$). In contrast, Prospective studies did not reach statistical significance, showing a pooled OR of 1.36 (95% CI: 0.96–1.91, $p = 0.045$) (Table 2).

3.3.14. By quality of studies

Both moderate and high-quality studies demonstrated consistent and statistically significant associations, with pooled ORs of 1.57 (95% CI: 1.27–1.95, $p < 0.001$) and 1.28 (95% CI: 1.20–1.37, $p < 0.001$), respectively (Table 2).

3.4. Heterogeneity and sensitivity analyses

Considerable heterogeneity was found across the included studies ($I^2 = 95.54\%$, $p < 0.001$), which justified the application of a random-effects model for pooling effect sizes. To assess the robustness of our results, sensitivity analyses were conducted by systematically removing studies with lower methodological quality and those exhibiting extreme effect estimates. The overall association between



Meta Analysis

Figure 2. Association between maternal anemia and preterm birth.

maternal anemia and preterm birth remained consistent, indicating that no single study unduly influenced the meta-analysis outcomes (Figure 3).

3.5. Publication bias

Funnel plot analysis showed an asymmetric distribution, which is indicative of publication bias. Egger's test detected statistically significant bias ($p=0.001$), suggesting potential asymmetry in the pooled results, while Begg's test did not reveal significant bias ($p=0.425$) (Figure 4).

5. Discussion

The present meta-analysis provides compelling evidence that maternal anemia significantly increases the risk of preterm birth, with a pooled OR of 1.51 (95% CI: 1.34–1.70, $p<0.001$). These findings

Table 2. Subgroup analysis of the association between maternal anemia and preterm birth based on continent, trimester of pregnancy, and study design.

Variable		Studies (N)	Heterogeneity		P-Value	95% CI	OR ³
			I ²	P-Value			
Continent	Africa	8	93.70	<0.001	0.091	0.89–4.40	1.98
	America	3	49.25	0.139	0.62	0.64–1.30	0.91
	Asia	38	96.62	<0.001	<0.001	1.22–1.40	1.30
	Europe	11	67.04	<0.001	0.001	1.08–1.41	1.24
Test for subgroup differences: $Q=5.185$, $df(Q)=3$, $p=0.159$							
Trimester	All	1	–	–	<0.001	1.12–1.19	1.16
	First	17	88.49	<0.001	0.007	1.03–1.22	1.12
	First-and-second	4	0	0.44	<0.001	1.87–4.16	2.79
	Second	8	62.19	0.01	0.246	0.94–1.24	1.08
	Second-and-third	1	–	–	0.050	0.04–1.00	0.20
	Third	29	92.65	<0.001	<0.001	1.42–1.91	1.65
Test for subgroup differences: $Q=45.64$, $df(Q)=5$, $p<0.001$							
Design	Prospective	5	87.60	<0.001	0.078	0.96–1.91	1.36
	Retrospective	55	95.73	<0.001	<0.001	1.20–1.35	1.27
Test for subgroup differences: $Q=0.123$, $df(Q)=1$, $p=0.726$							
Quality	High	37	96.79	<0.001	<0.001	1.20–1.37	1.28
	Moderate	23	88.62	<0.001	<0.001	1.27–1.95	1.57
Test for subgroup differences: $Q=3.118$, $df(Q)=1$, $p=0.077$							

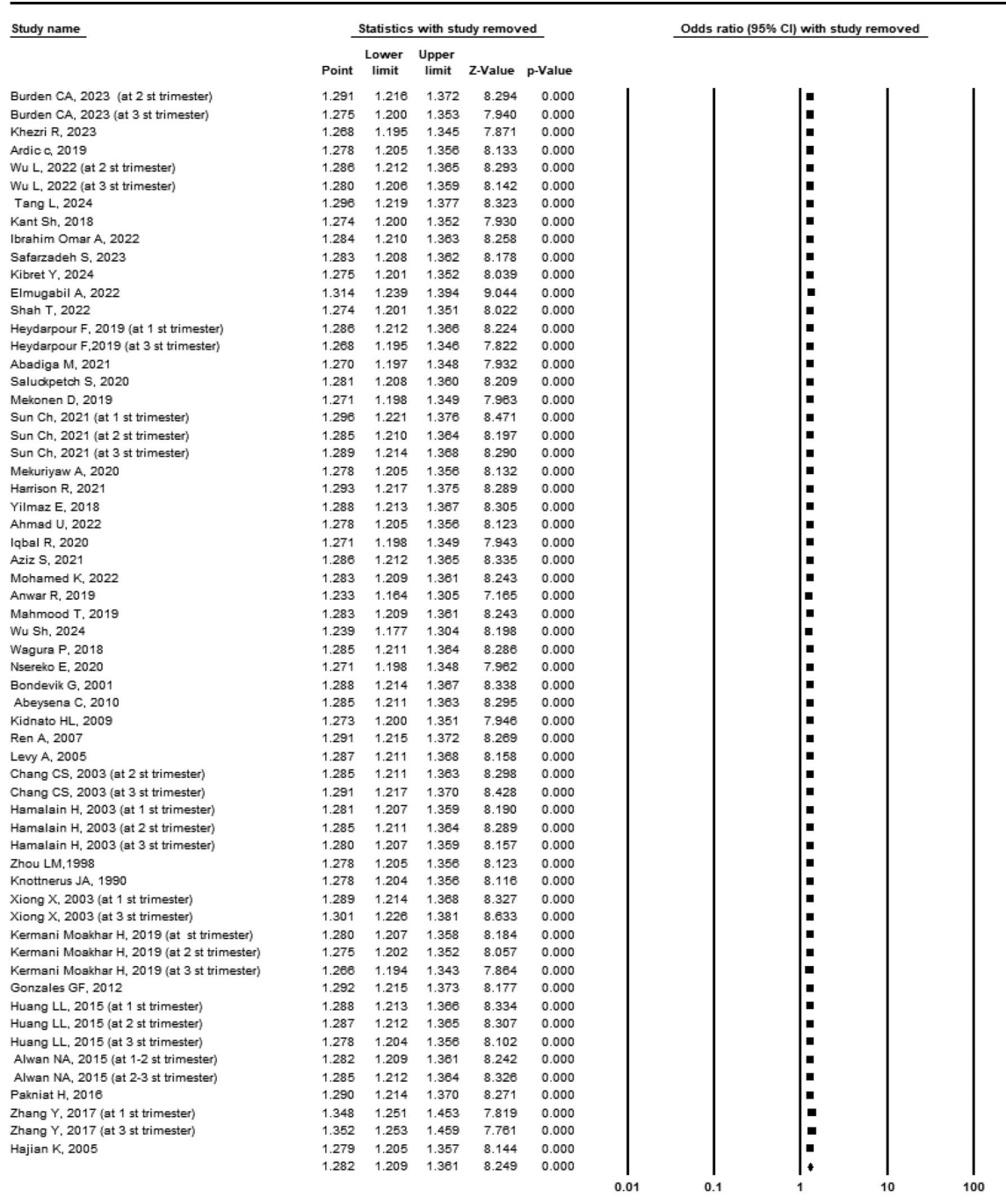
N: number; CI: confidence interval; OR: odds ratio.

are consistent with recent literature [51,59–61] which posits that maternal anemia contributes to preterm delivery primarily through mechanisms such as reduced oxygen-carrying capacity, fetal hypoxia, placental insufficiency, and systemic inflammation [61]. A 2018 meta-analysis investigating the association between maternal anemia and preterm birth, based on 18 studies, reported a significant OR of 1.56 (95% CI: 1.25–1.95). This analysis indicated that the association was significant in the first trimester but not in the second or third [3]. Our findings strongly corroborate this association, yielding a narrower 95% CI. However, in contrast to the previous study, our analysis revealed no significant association during the first trimester; instead, significance was limited to the third trimester. We attribute this to the larger number of studies included in our meta-analysis, which likely provides a more accurate estimate.

The subgroup analyses by geographic region reveal intriguing disparities in the strength of this association. The strongest effects were observed in Asian populations (OR = 1.30) followed by those in the Europe (OR = 1.24), while the association was weaker and statistically non-significant in some African and American cohorts. These regional variations likely reflect differences in socioeconomic factors, healthcare infrastructure, nutritional status, prevalence of infectious diseases, and genetic predispositions that influence anemia severity and pregnancy risks [62,63]. For example, in low-resource settings, where iron deficiency and malaria are more prevalent, anemia may have a more profound impact on fetal development and gestational duration.

Regarding study design, no significant difference was observed between the results of retrospective and prospective studies, underscoring the robustness and reliability of the evidence. The methodological consistency observed among the studies enhances the credibility of a causal link between maternal anemia and preterm birth, while reducing the likelihood that the findings are due to confounding variables or biases specific to individual studies. However, retrospective studies face limitations including the possibility of bias, limited access to accurate and complete data, and the inability to control variables [61,64].

A particularly noteworthy finding of this analysis is the trimester-specific effect of anemia on preterm birth risk. Anemia diagnosed during the third trimester was significantly associated with a higher risk (OR = 1.51), whereas anemia in the first and second trimesters was not statistically significant. This temporal pattern may reflect the critical importance of adequate maternal hemoglobin levels in late pregnancy, a period marked by rapid fetal growth and heightened oxygen demands [65]. The compromised oxygen delivery and placental function in the third trimester likely precipitate preterm labor and delivery. These findings align with recent evidence emphasizing the importance of the timing of anemia during pregnancy in relation to adverse birth outcomes [4,61].



Meta Analysis

Figure 3. Sensitivity analysis of the association between maternal anemia and preterm birth.

Despite substantial heterogeneity among included studies ($I^2 = 95.54\%$), the use of a random-effects model was appropriate to account for variability in study populations and designs. Sensitivity analyses further confirmed the stability of the findings, demonstrating that exclusion of studies with lower methodological quality or extreme effect sizes did not alter the overall conclusions. The consistency of these findings strengthens the reliability of the meta-analysis outcomes.

Furthermore, the assessment of publication bias using funnel plots and formal tests (Egger's and Begg's) did not reveal significant bias, indicating that the pooled effect estimates are unlikely to be

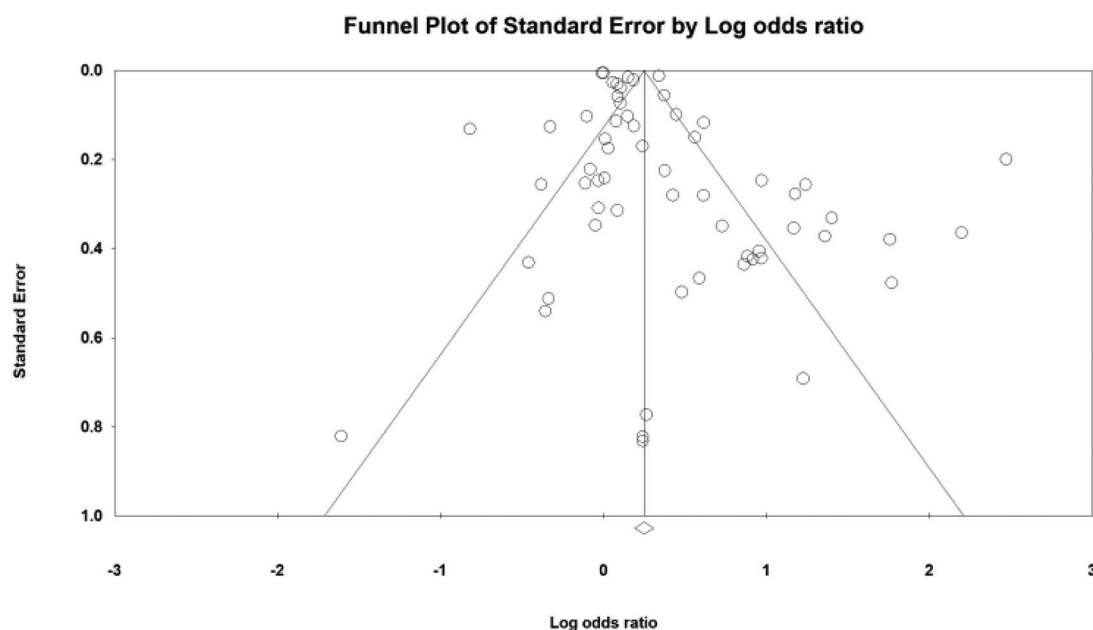


Figure 4. Publication bias among studies included in the meta-analysis.

influenced by selective publication. This evidence supports the conclusion that maternal anemia plays a significant role in increasing the risk of preterm delivery.

6. Limitations

Despite the robust methodology employed, including adherence to PRISMA and MOOSE guidelines, and comprehensive subgroup and sensitivity analyses, our meta-analysis has several limitations that warrant consideration. Firstly, a substantial level of heterogeneity was observed across the included studies ($I^2 = 92.95\%$). This extreme variability likely stems from significant clinical and methodological inconsistencies, including divergent definitions of anemia, varying hemoglobin thresholds, and differing timings of anemia assessment. Secondly, all included studies were observational in design, which precludes the establishment of definitive causality. Despite adjustments for confounders in individual studies, the possibility of residual or unmeasured confounding factors influencing the observed association cannot be entirely ruled out. Thirdly, our search strategy was limited to studies published in English, potentially introducing a language bias by excluding relevant research published in other languages. This might affect the comprehensiveness of our findings and could lead to an incomplete representation of the global evidence. Lastly, while subgroup analyses provided valuable insights into the sources of heterogeneity, the findings from these analyses, particularly those with a smaller number of studies, should be interpreted with caution. Future research, especially large-scale prospective cohort studies with standardized definitions and comprehensive adjustment for confounders, is needed to further elucidate the intricate relationship between maternal anemia and preterm birth.

7. Conclusions

This meta-analysis indicates that maternal anemia is a significant risk factor for preterm birth, with the most pronounced effect observed when anemia occurs during the third trimester. These results highlight the critical importance of early detection and targeted treatment of anemia during pregnancy to prevent adverse perinatal outcomes. While substantial heterogeneity was observed among included studies, the overall association underscores the clinical relevance of maternal hemoglobin monitoring as part of comprehensive prenatal care. Future research should focus on elucidating the pathophysiological pathways linking anemia to preterm labor and evaluating the efficacy of trimester-specific therapeutic strategies. Ultimately, proactive management of maternal anemia holds promise as a pivotal strategy to reduce global preterm birth rates and improve both maternal and neonatal health outcomes.

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Author contributions

CRediT: **Parviz Karimi**: Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Zahra Nourollahi**: Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing; **Mohammad Mehdi Nourollahi**: Conceptualization, Data curation, Investigation, Methodology, Resources, Validation, Writing – original draft, Writing – review & editing; **Milad Azami**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

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Data availability statement

All data generated or analyzed during this study are included in the published article and its supplementary materials.

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